

Sienna White

Berkeley, California | siennaw@berkeley.edu | siennarwhite.github.io | github.com/siennarwhite

Research Experience

University of California, Berkeley Aug 2022 – Present

Ph.D. Student, Environmental Engineering

- Working with Professor Tina Chow to develop data assimilation applications for high-resolution wild-fire smoke models.

San Francisco Estuary Institute Sep 2020 – Dec 2022

Environmental Analyst

- Developed biogeochemical and hydrodynamic numerical models to inform management decisions for the greater Bay Area.

Technische Universiteit Delft (Fulbright Fellow) Sep 2019 – Jun 2020

Researcher in River Morphology

- Quantified morphological change from fixed river beds in the Waal with Rijkswaterstaat and TU Delft.

Deltares Jun 2018 – Sep 2018

Rivers Department

- Processed bathymetry data and experimented with methods for parametrizing bedform drag.

Environmental Fluid Mechanics Lab, Stanford May 2016 – Sep 2019

Sediment Transport Research

- Performed statistical analysis of sedimentation budgets for San Francisco Bay; supervised by Oliver Fringer and Joe Adelson.

Education

Stanford University Sep 2015 – Jun 2019

B.S. with Honors, Environmental Engineering

- Thesis: *A Numerical Model for Sediment Flux and Flocculation in San Francisco Bay.*

Skills

Python (incl. geospatial analysis), MATLAB, QGIS, numerical hydraulic modeling (DFlow-FM, DELWAQ), Linux/Unix, Git, UAV pilot license, technical & long-form writing.

Publications & Presentations

Technical Reports

- S.R. White, et al. “Delta–Suisun Biogeochemical Model: Calibration and Validation.” San Francisco Estuary Institute, 2021.
- S.R. White, et al. “Nutrients in the Northern San Francisco Estuary.” San Francisco Estuary Institute, 2021.

Conference Presentations

- S.R. White, A. Blom. “Quantifying sediment transport...” Netherlands Centre for River Studies, 2020.

- S.R. White, O.B. Fringer. “Using D-Water Quality and D-Flow FM...” Delta Science and Software Days, 2019.

Awards & Scholarships

- Firestone Medal, top 10% of honors theses at Stanford (2019)
- 1st Place, Creative Nonfiction Prize, Stanford (2017)
- Human Cities Urban Innovation Contest Finalist (2016)
- US Presidential Scholar (2015)

Other Work Experience

- Teaching Assistant, Stanford University (2016–2019)
- Senior Producer, Stanford Storytelling Project (2016–2019)

References

Available upon request.